

ALANA DENG

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Education

Simon Fraser University

PhD in Statistics

Burnaby, BC, Canada

Sept 2025 - Aug 2029 (Expected)

Western University

MSc in Computer Science (Thesis)

London, Ontario, Canada

Sept 2023 - Aug 2025

- **Coursework:** AI II (CS 9542), Theoretical Machine Learning (CS 9875), Reinforcement Learning (CS 9670), AI for Genome Science (BIOCHEM 9551)
- **Grade:** 95/100 (A+)

Jinan University (JNU)

BEng in Computer Science and Technology

Guangzhou, China

Sept 2019 - July 2023

- **Selected Coursework:** Linear Algebra, Probability and Statistics, Algorithm Design and Analysis, Machine Learning
- **Grade:** 89/100 (top 3 in class 2023)

Experience

Department of Computer Science, Western University

Graduate Research Student

London, Ontario, Canada

Sept. 2023 – Present

- **Project:** Deep Learning Framework for Interaction Prediction in Biological Networks
 - * Evaluated multiple graph representation techniques, including *DeepWalk*, *Node2Vec*, *Struc2Vec*, and *BiNE*, for use in transductive link prediction.
 - * Applied Graph Neural Networks (GNNs) such as *GCNs*, *GATs*, Graph Transformer, and Deep Graph Infomax (DGI) for multiplex biological networks.
 - * Developing a zero-shot multiplex interaction prediction framework for drug discovery that includes:
 - a domain-specific Large Language Model (LLM) module for biochemical sequence embedding;
 - a Unified Graph Tokenizer for topology-aware, multiplex-aware embedding generation;
 - a Context-Aware Enhancement (CAE) module to refine inter-layer multiplex representations;
 - a Mixture of Experts classifier for multi-label link prediction;
 - a hybrid knowledge distillation and self-adaptive inference strategy for robust zero-shot prediction.
- **Project:** Alignment-Free Machine Learning for HIV-1 Subtype Classification
 - * Applied alignment-free DNA encoding techniques: *Ordinal Encoding*, *k-mer*, *Sub-sequence Natural Vector*, and NLP-inspired methods (*Word2Vec*, TF-IDF).
 - * Used UMAP/PCA for dimensionality reduction and SMOTE for handling class imbalance.
 - * Built classifiers using *Logistic Regression*, *Naive Bayes*, *XGBoost*, and *SVM*.

Teaching Assistant

Sept. 2023 – Present

- CS3340B – Analysis of Algorithms I
- CS3331A – Foundations of Computer Science (Automata, Computability)
- CS1027B – Fundamentals II (Java Object-Oriented Programming)
- CS1026A – Fundamentals I (Intro to Python Programming)

Department of Mathematics and Statistics, University of Victoria

Research Intern

Victoria, BC, Canada

June 2022 – Sept. 2022

- **Project:** Machine Learning for Survival Data
 - * Explored Accelerated Failure Time (AFT) models with custom simulation baselines.
 - * Developed hybrid models combining AFT with *XGBoost*, *Lasso*, and *Elastic Net*.
 - * Performed censored regression on synthetic and real clinical datasets.
 - * Participated in the 2022 Statistics Conference on Genomics, Pharma, and Health Data Science.

Department of Computer Science, Jinan University

Undergraduate Research Student

Guangzhou, China

May 2021 – June 2023

- **Project:** AI-Assisted CHD Recurrence Risk Prediction

- * Preprocessed clinical data (e.g., imputation, correlation testing, feature selection, abnormal data handling).
- * Built survival models: *Kaplan-Meier*, *CoxPH*, *DeepSurvival*, and neural AFT models.
- * Enhanced non-parametric AFT models using hybrid rank- and error-based loss in PyTorch.
- * Integrated DAG-GNN for multimodal health data and structure learning.
- **Project:** Time Series Forecasting for Tax Service Demand
 - * Implemented *ARIMA*, *DeepAR*, *Transformer*, and *Informer* models.
 - * Applied survival analysis for predicting service waiting times.
- **Project:** Investigating Goji Berry's Effects on Retinal Degeneration
 - * Conducted behavioral experiments (e.g., Morris Water Maze) in collaboration with medical researchers.
 - * Performed statistical evaluation using Student's t-test and z-test.

Publications

- Wade, K. E., Chen, L., Deng, C., Zhou, G., & Hu, P. (2024). Investigating alignment-free machine learning methods for HIV-1 subtype classification. *Bioinformatics Advances*, 4(1), vbae108.
- Hadipour, H., Li, Y. Y., Sun, Y., Deng, C., Lac, L., Davis, R., ... & Hu, P. (2025). GraphBAN: An inductive graph-based approach for enhanced prediction of compound-protein interactions. *Nature Communications*, 16(1), 2541.

Honors & Awards

Western Graduate Research Scholarship	2023, 2024, 2025
Mitacs Globalink Fellowship Award	2023
JNU Outstanding Graduates Award	2023
Jinan University Outstanding Graduate Thesis Award	2023
China National Scholarship	2022
Mitacs Globalink Research Internship Award	2022
JNU Outstanding Students Award	2020, 2021, 2022

Technical Skills

Languages: English (Professional Working Efficiency, IELTS 8/9), Chinese (Native)
Programming: Python, R, Matlab, Java, C, C++, Markdown, SQL, Front-End Development
Technologies/Frameworks: Linux/Unix, Deep learning (PyTorch, TensorFlow, etc.), Machine Learning (Scikit-Learn, XGBoost, etc.), Survival Analysis, Reinforcement Learning (OpenAI Gym)